Reflect with us - Week 3

This assignment will not be graded and is only for practice.

1 point

Choose the set of correct options.

- [ ] Any subset of any linearly independent set of vectors is linearly independent.
- [ ] Any subset of any linearly dependent set of vectors is linearly dependent.
- [ ] Any subset of any linearly dependent set of vectors is linearly independent.
- [ ]

Any set of 33 vectors in R^3 R3 is linearly independent.

- [ ]

Any set of 44 vectors in R^3 R3 is linearly dependent.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Any subset of any linearly independent set of vectors is linearly independent.
Any set of 44 vectors in R^3 R3 is linearly dependent.

Recall the definition of linearly dependent and independent:
Recall the definition of linearly dependent and independent:

Statement 1) Definition of linear dependence: A set of vectors $\{v_1, v_2, \ldots v_n\}$ {v1, v2, … vn} in V V is said to be linearly dependent if there exist scalars $a_1, a_2, \ldots a_n \in R$ a1, a2, … an ∈ R, not all zero such that $a_1 v_1 + a_2 v_2 + \ldots + a_n v_n = 0$ a1v1 + a2v2 + … + anvn = 0.

Statement 2) Definition of linear independence: A set of vectors $\{v_1, v_2, \ldots v_n\}$ {v1, v2, … vn} in V V is said to be linearly independent if $a_1 v_1 + a_2 v_2 + \ldots + a_n v_n = 0$ a1v1 + a2v2 + … + anvn = 0 implies that $a_i = 0$ ai = 0 for $i = 1, 2, \ldots n$ i = 1, 2, … n.

Statement 3) If SS is a set containing zero vector, then the set SS is a linearly dependent set.

Statement 4) Let SS be a subset of a vector space VV . If an element of SS is a scalar multiple of another element from the set SS, then SS is a linearly dependent set.

Statement 5) If SS is linearly dependent, then there exists $v \in S$ v ∈ S, such that vv can be written as a linear combination of other vectors in SS and vice versa.

Discussion on Option 1:Discussion on Option 1: Let $S = \{v_1, v_2, \ldots, v_n\}$ S = {v1, v2, … , vn} be a linearly independent set of vectors and let S' S′ be a subset of SS.

Claim:Claim: S' S′ is linearly independent.

1 point

What we have to assume for S' S′ if we want to use the method of contradiction?

- ( )

Let us assume that S' S′ is linearly dependent.

- ( )

Let us assume that S' S′ is linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Let us assume that S' S′ is linearly dependent.

If S' S′ is linearly dependent, then there exists $v \in S'$ v ∈ S′, such that vv can be written as a combination of other vectors in S' S′.

Which of the above statements we are using here?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 5
1 point

$v \in S'$ implies $v \in S$, as $S' \subseteq S$. Those vectors of whose linear combination is giving v are also in S due to the same argument.
1 point
Again from statement 5, can we conclude that S is linearly dependent?
○ Yes.
○ No.
No, the answer is incorrect.
Score: 0
Accepted Answers:
Yes.

But S was given to be a linearly independent set, hence we arrive at a contradiction. So our claim is proved.
Another method: Let $S' = \{v_{i_1}, v_{i_2}, \ldots, v_{i_k}\} \subseteq S$. Consider the equation

$$a_1 v_{i_1} + a_2 v_{i_2} + \ldots + a_k v_{i_k} = 0$$

1 point
Can we conclude each $a_i = 0$ from here?
○ Yes.
○ No.
No, the answer is incorrect.
Score: 0
Accepted Answers:
Yes.

As all the v_{i_j} are also in S, the given equation is a linear combination of vectors from S. Now as S is linear independent, $a_i = 0$ for each $i \in \{1, 2, \ldots, k\}$. Hence S' is linearly independent.
$\because$ Now try Question 7 and 8 of activity 3.3

Discussion on Option 2: Let us discuss an example. Let $S = \{(1,0,0), (0,1,0), (1,1,0)\}$ be a subset of R^3.
1 point
Is S linearly dependent?
○ Yes.
○ No.
No, the answer is incorrect.
Score: 0
Feedback:

Hint: (1, 1, 0) = (1, 0, 0) + (0, 1, 0)

Accepted Answers:

Yes.

1 point

Can you find a linearly independent subset of SS?

◯ Yes.

◯ No.

No, the answer is incorrect.

Score: 0

Feedback:

Hint: Check that any proper subset of SS of cardinality 1 or 2 is linearly independent subset of SS.

Accepted Answers:

Yes.

Consider $S' = \{(1, 0, 0), (0, 1, 0)\}$S′ = {(1, 0, 0), (0, 1, 0)}.

1 point

Is $S' \subseteq S$S′ ⊆ S?

◯ Yes.

◯ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

1 point

Is S' S′ linearly independent?

◯ Yes.

◯ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

Hence Option 2 is not correct.

Discussion on Option 3:Discussion on Option 3: Here also we construct an example. Let $S = \{(1, 0, 0), (2, 0, 0), (3, 0, 0)\}$S = {(1, 0, 0), (2, 0, 0), (3, 0, 0)}.

1 point

Is SS linearly dependent?

◯ Yes.

◯ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

Consider $S' = \{(1, 0, 0), (2, 0, 0)\}$S′ = {(1, 0, 0), (2, 0, 0)}.

1 point

Is $S' \subseteq S$ S′ ⊆ S?

◯ Yes.

◯ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

1 point

Is S' S′ linearly independent?

◯ Yes.

◯ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

No.

Hence Option 3 is not correct.

Discussion on Option 4:Discussion on Option 4: Consider the same set $S = \{(1,0,0),(2,0,0),(3,0,0)\}$ S = {(1, 0, 0), (2, 0, 0), (3, 0, 0)} that we have discussed previously.

1 point

Is SS linearly independent?

◯ Yes.

◯ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

No.

1 point

What can we conclude about Option 4 from this?

◯ Option 4 is correct.

◯ Option 4 is not correct.

◯ Nothing can be concluded about Option 4 from here.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Option 4 is not correct.

· · Now try Question 5 and 8 of activity 3.4Now try Question 5 and 8 of activity 3.4

Discussion on Option 5:Discussion on Option 5: Let $S = \{v_1, v_2, v_3, v_4\}$S = {v1, v2, v3, v4} be a set of 4 vectors in R^3 R3. Let $v_1 = (x_1, y_1, z_1)$v1 = (x1, y1, z1), $v_2 = (x_2, y_2, z_2)$v2 = (x2, y2, z2), $v_3 = (x_3, y_3, z_3)$v3 = (x3, y3, z3), $v_4 = (x_4, y_4, z_4)$v4 = (x4, y4, z4). Consider the following equation,

$$a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = 0$$ a1v1 + a2v2 + a3v3 + a4v4 = 0

i.e.,

$$a_1(x_1, y_1, z_1) + a_2(x_2, y_2, z_2) + a_3(x_3, y_3, z_3) + a_4(x_4, y_4, z_4) = (0, 0, 0)$$
a1(x1, y1, z1) + a2(x2, y2, z2) + a3(x3, y3, z3) + a4(x4, y4, z4) = (0, 0, 0)

i.e.,

$$\begin{aligned} a_1 x_1 + a_2 x_2 + a_3 x_3 + a_4 x_4 &= 0 \\ a_1 y_1 + a_2 y_2 + a_3 y_3 + a_4 y_4 &= 0 \\ a_1 z_1 + a_2 z_2 + a_3 z_3 + a_4 z_4 &= 0 \end{aligned}$$
a1x1 + a2x2 + a3x3 + a4x4a1y1 + a2y2 + a3y3 + a4y4a1z1 + a2z2 + a3z3 + a4z4 = 0 = 0 = 0

The matrix representation of this system of linear equations is given by

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \\ z_1 & z_2 & z_3 & z_4 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$
x1y1z1x2y2z2x3y3z3x4y4z4 a1a2a3a4 = 000

We want to find the solution for the vector $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$ a1a2a3a4 for this given homogeneous system of linear equations.

1 point
What can we say about the solutions of this system of linear equations?
- [ ] The solution is unique.
- [ ] There are infinitely many solutions.
- [ ] There must exist a non-zero solution.
- [ ]

00 is the only solution.

No, the answer is incorrect.
Score: 0
Feedback:
Hint: Try to compute the maximum number of dependent variables you can get from the row reduced echelon form of the above matrix.
Accepted Answers:
There are infinitely many solutions.
There must exist a non-zero solution.

1 point
Is SS linearly independent?
- ○ Yes.
- ○ No.

No, the answer is incorrect.
Score: 0
Accepted Answers:
No.

Hence Option 5 is correct.

· ·This will help you to solve Question number 4 of the graded assignment.

This will help you to solve Question number 4 of the graded assignment.

Check Answers

Your score is: 0/16